

Monthly Intelligence Report

Edition 01 — The Savinja Valley Corridor

Where the energy transition is outrunning the grid. Inaugural edition · July 2026 · SSI v4.0.2

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SECTION A

Executive Summary

SUBSTATIONS SCORED

158

21 EHV · 137 HV

GRID LINES MAPPED

~860

~21,000 km coverage

MC SIMULATIONS

1.58 M

10,000 per substation

PUBLIC DATA SOURCES

30+

95 variables · zero proprietary

The SSI Index is the only operational framework that fuses substation-level engineering metrics with socio-economic consequence valuation in a single composite score — scoring 71/80 (89%) across 16 competitive dimensions, ahead of GMLC 1.1 (48/80), EPRI (40/80), and academic MCDA frameworks (41/80). Its engine processes 32,000+ source data points per run, executes 1.58 million Monte Carlo simulations through a 20×20 Gaussian copula correlation matrix, and performs 36 million arithmetic operations — producing 9,480 output data points with full uncertainty quantification across 158 substations and ~860 grid lines spanning ~21,000 km.

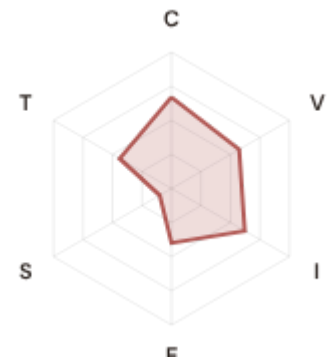
Operated as a non-profit through a dedicated foundation, the SSI Index serves the public interest with zero dependency on proprietary data. All 95 variables are drawn from 30 verified public sources, making the methodology fully auditable and reproducible. Initially covering Slovenia's entire transmission and distribution grid, the Index is progressively being rolled out to all OECD countries — applying the same silo-breaking approach: fusing grid risk with societal exposure at asset-level resolution wherever open data permits.

Substation in Focus

Trbonje-Trbul

SIO33, SIO33 · 66 kV

0.532 **High** **0.092** **98th**
R_FINAL CLASSIFICATION CI WIDTH FLEET PERCENTILE



COMPONENTS

C Continuity: 0.670 / 0.30 (223%) ▲
I Infrastructure: 0.621 / 0.25 (248%)
S Saturation: 0.100 / 0.20 (50%)
V Voltage: 0.574 / 0.10 (574%)
E Economic: 0.399 / 0.10 (399%)
T Transition: 0.440 / 0.05 (880%)

MODIFIERS

R3 Consequence: 0.924 ↓ dampens
R4 Graph Criticality: 1.001 → neutral
R6a Restoration: 1.078 ↑ amplifies
R6b Seismic: 1.048 ↑ amplifies
R7 Digital Readiness: 1.026 ↑ amplifies

This month's spotlight — automatically selected for June 2026 — is **Trbonje-Trbul** in SIO33, SIO33. With an R_final of 0.532 (High band, 98th fleet percentile), its risk profile is dominated by the T (Transition) component at 880% of its weight ceiling, followed by V (Voltage) at 574%. Modifiers collectively shift R_base by 14.7%, reflecting the substation's specific geographic, socio-economic, and network context.

SECTION B — RECURRING MODULE

Cyber & Economic Exposure Monitor

Why this section exists: Traditional grid risk frameworks treat cybersecurity and economic vulnerability as separate domains. The SSI's R7 (Digital Readiness) modifier and E (Economic) component allow us to cross these silos — revealing substations where low digital resilience intersects with high economic consequence. This is precisely the anticipatory intelligence that Slovenia's Digital Slovenia 2030 / NEPN digitalisation planning requires.

CYBER HIGH EXPOSURE

0

0.0% of fleet

BLIND SPOTS

1

High/Critical + R7 < median

AVG R7 MODIFIER

0.9796

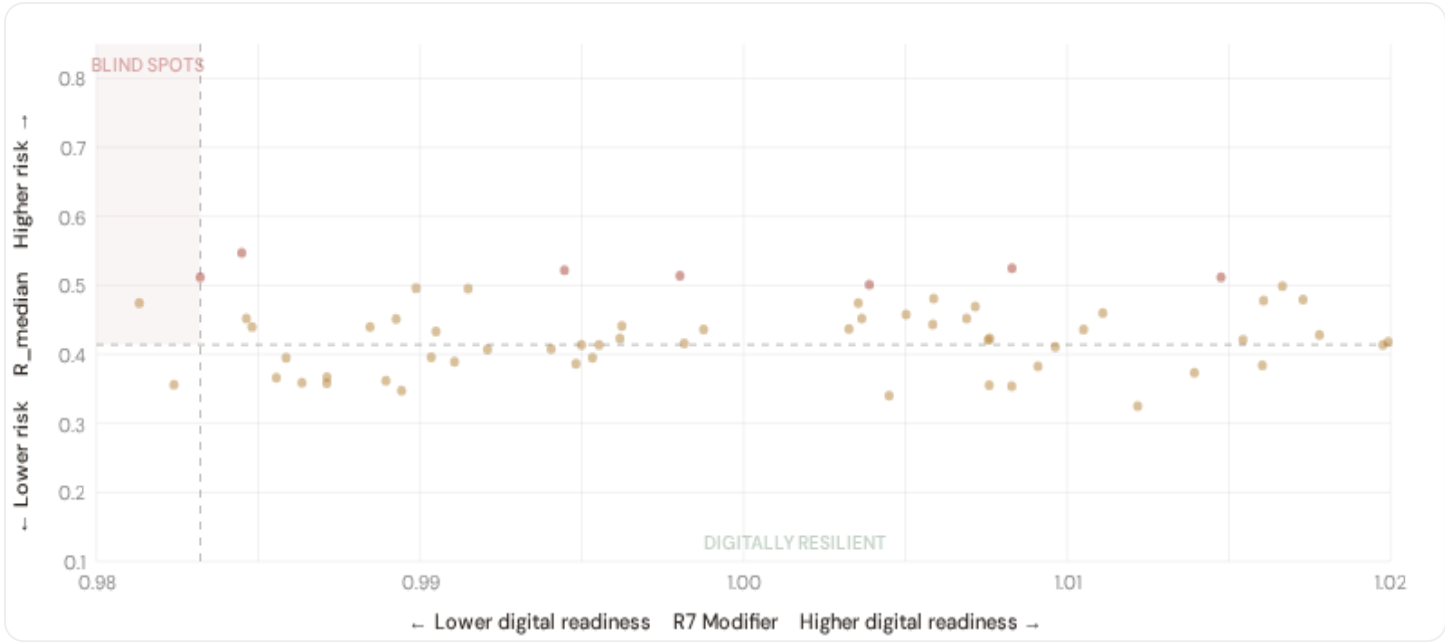
Range [0.99, 1.05]

HIGH-CONSEQUENCE SUBS

40

B.1 Cyber-Physical Risk Matrix

Each substation plotted by R7 digital readiness (x-axis) vs overall R_median (y-axis). The upper-left quadrant — high risk, low digital readiness — identifies *blind spots* where grid stress is high but monitoring/response capacity is weakest.

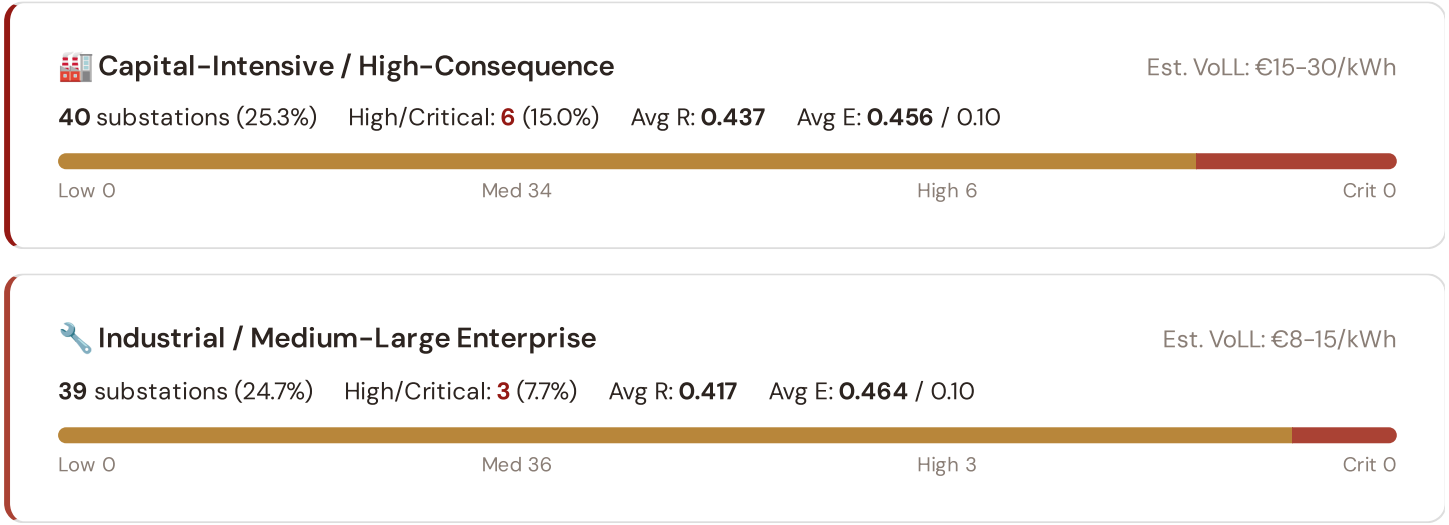


● Low ● Medium ● High ● Critical Dashed lines = fleet medians

The cyber-physical matrix identifies **1 blind-spot substations** — scoring High or Critical on overall risk while sitting below the fleet median for digital readiness ($R7 < 0.9832$). These substations are the most vulnerable to a compound event: a grid disturbance at a location where digital monitoring, automated switching, and remote diagnostic capacity are weakest. The blind spots concentrate in **SI043 (1)**. Across the full fleet, 0 substations (0.0%) carry a HIGH cyber-exposure classification, while 0 (0.0%) are in the LOW tier.

B.2 Economic Impact by Business Fabric

Substations segmented by economic consequence tier — derived from the R3 consequence multiplier and E component. Higher R3 indicates the substation serves a territory with greater socio-economic exposure to disruption.



Commercial / Mixed

Est. VoLL: €3–8/kWh

40 substations (25.3%) High/Critical: 1 (2.5%) Avg R: 0.403 Avg E: 0.470 / 0.10

Low 0 Med 39 High 1 Crit 0

Light-Rural / Agricultural

Est. VoLL: €1–3/kWh

39 substations (24.7%) High/Critical: 3 (7.7%) Avg R: 0.393 Avg E: 0.473 / 0.10

Low 0 Med 36 High 3 Crit 0

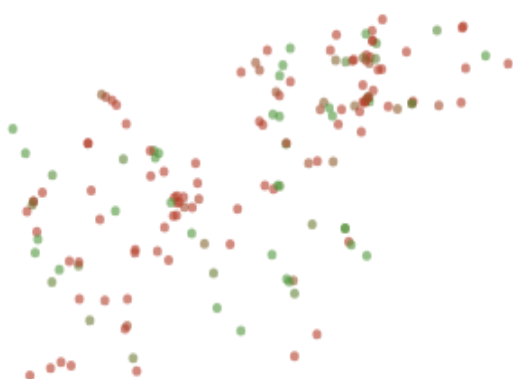
Value of Lost Load (VoLL) context: ACER's 2023 estimate places average VoLL at €13.1/kWh for industrial customers and €3.2/kWh for residential. The SSI's R3 consequence multiplier allows us to identify which substations serve the most economically exposed territories. **6 substations** combine capital-intensive economic fabric ($R3 \geq 1.05$) with High or Critical risk classification — representing the highest VoLL-weighted exposure in the fleet.

The capital-intensive tier concentrates in **SIO32 (8), SIO41 (7), SIO43 (5)** — regions where industrial corridors depend on uninterrupted power for continuous processes. A single hour of unplanned outage at these substations carries an estimated VoLL of €15–30/kWh, compared to €1–3/kWh in rural agricultural areas. The fleet-wide average energy poverty rate is 5.4%, but this masks sharp regional variation: Southern regions combine higher energy poverty with higher grid risk, creating a double vulnerability that the E component captures. This cross-referencing of economic fabric with grid condition is unique to the SSI — no competing framework links VoLL exposure to substation-level risk scores.

B.3 NUTS-3 Region Digital Readiness Ranking

NUTS-3 Regions ranked by average R7 modifier — lower values indicate weaker digital infrastructure (DESI sub-dimensions). Cross-referenced with average R_median to identify NUTS-3 regions combining digital exposure with high grid stress.

R7 Digital Readiness
Low High



WORST DIGITAL READINESS — TOP 15 NUTS-3 REGIONS

Longer bar = higher R7 = better digital infrastructure.

1	SIO44 ▲ high R	0.9590
2	SIO31	0.9652
3	SIO41	0.9720
4	SIO32	0.9722
5	SIO38 ▲ high R	0.9735
6	SIO42	0.9807
7	SIO33 ▲ high R	0.9843
8	SIO43 ▲ high R	0.9846
9	SIO34 ▲ high R	0.9868
10	SIO35 ▲ high R	0.9877
11	SIO36 ▲ high R	1.0026
12	SIO37 ▲ high R	1.0071

NUTS-3-level analysis reveals a clear digital divide mirroring the broader DESI pattern. **SIO44** has the lowest average R7 (0.9590), while **SIO37** leads at 1.0071 — a spread of 0.0481 across the R7 range. **2 nuts-3 regions** combine below-median digital readiness with above-median grid risk, making them priority targets for digitalisation investment. The R7 modifier is intentionally narrow to reflect the current limitations of open DESI data — as NIS2 compliance reporting matures, future editions will expand R7's dynamic range and incorporate operational technology (OT) security indicators.

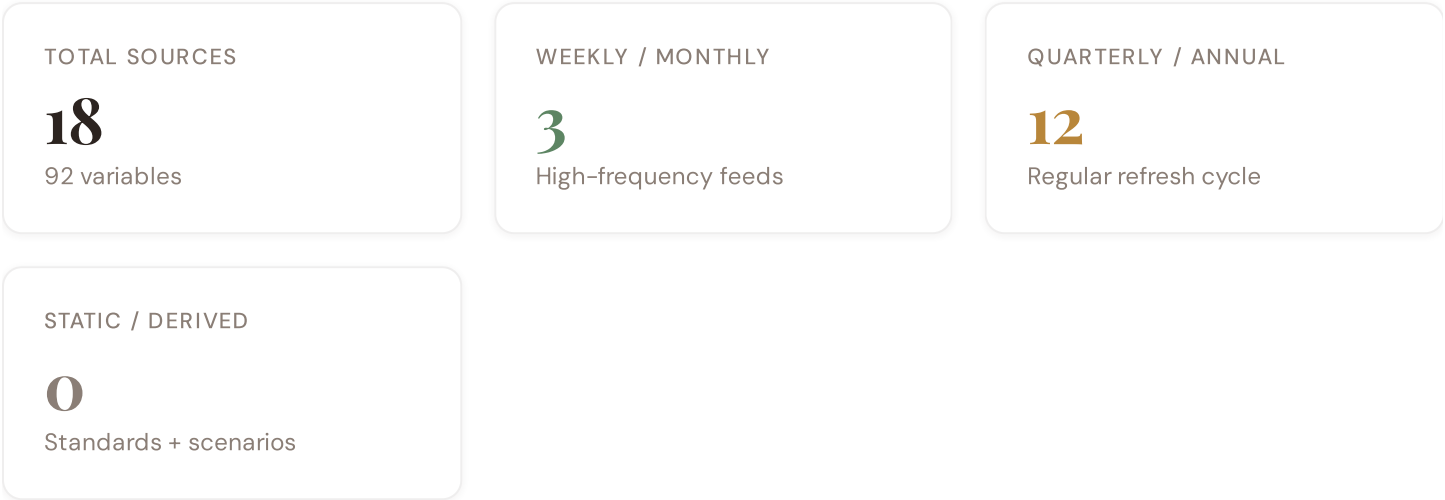
B.4 Monthly Pulse

Inaugural edition — Baseline Snapshot (June 2026). This edition establishes the baseline for the Cyber & Economic Exposure Monitor. Key reference points: fleet average R7 = 0.9796, median = 0.9832; 0 substations classified HIGH cyber-exposure; 1 blind-spot substations (High/Critical risk + below-median R7); 0 Critical-band substations with below-median digital readiness. Future editions will track deltas against this baseline.

SECTION C

Data Refresh & Changelog

The SSI v4.0.2 draws on **18 verified public data sources** feeding 92 variables across 6 components and 5 modifiers. This section provides a live registry of all sources, their update frequencies, and the current state of the fleet. Transparency in data vintage is central to the SSI's credibility.



Data Source Registry — June 2026

d01_eles	ELES — Elektro–Slovenija (TSO)	DAILY	C2,C3,I1,I2,I3,R4	14 vars
d03_arso_seismic	ARSO + ZAG-EQS — Seismology	CONTINUOUS	R6_seismic	4 vars
d03b_drsv	DRSV — Direkcija za vode (floods)	MONTHLY	R6c_flood	6 vars
d05_osm	OSM Overpass — grid topology	MONTHLY	I1,I2,I3,R4	8 vars
d06_arso_met	ARSO — Meteorology + climate	CONTINUOUS	R6_storm,R5_corro...	5 vars
d07_copernicus	Copernicus ERA5 + CMIP6	MONTHLY	R6_climate	4 vars
d09_si_cert	SI-CERT — National CERT (1995)	CONTINUOUS	R7_cyber	5 vars
d02b_dso	5 Elektro DSOs (Lj/Mb/Ce/Pr/Gr)	QUARTERLY	C1,V1,V2,I2,I3	18 vars
d04_surs	SURS — Statistični urad RS	QUARTERLY	E1,E2,S1,S2,S3	16 vars
d04b_bs	Banka Slovenije — central bank	QUARTERLY	E1,E2	4 vars
d08_ursjv	URSJV — Nuclear safety (Krško)	QUARTERLY	R6_seismic,I3	3 vars
d12_nek_krsko	NEK Krško — SI/HR joint operator	QUARTERLY	I1,I3,R6_seismic	4 vars
d02_agen_rs	AGEN-RS — Agencija za energijo	ANNUAL	C1,C2,V1,V2,T1	12 vars
d10_entsoe	ENTSO-E TYNDP + Transparency	ANNUAL	I1,I2,T1	6 vars
d11_eurostat	Eurostat — EU-27 benchmarks	ANNUAL	E1,S1,T1	8 vars
d13_hse	HSE — Hydro + lignite generation	ANNUAL	I1,T1	5 vars
d14_gen_energija	GEN Energija — Krško 50% + Brestanica	ANNUAL	I1,T1	3 vars
d15_iea_oecd	IEA + OECD — energy benchmarks	ANNUAL	T1,E1	6 vars

C.1 Update Frequency Timeline

EXPECTED UPDATE WINDOWS

Monthly (3 sources)

Grid load + procurement data

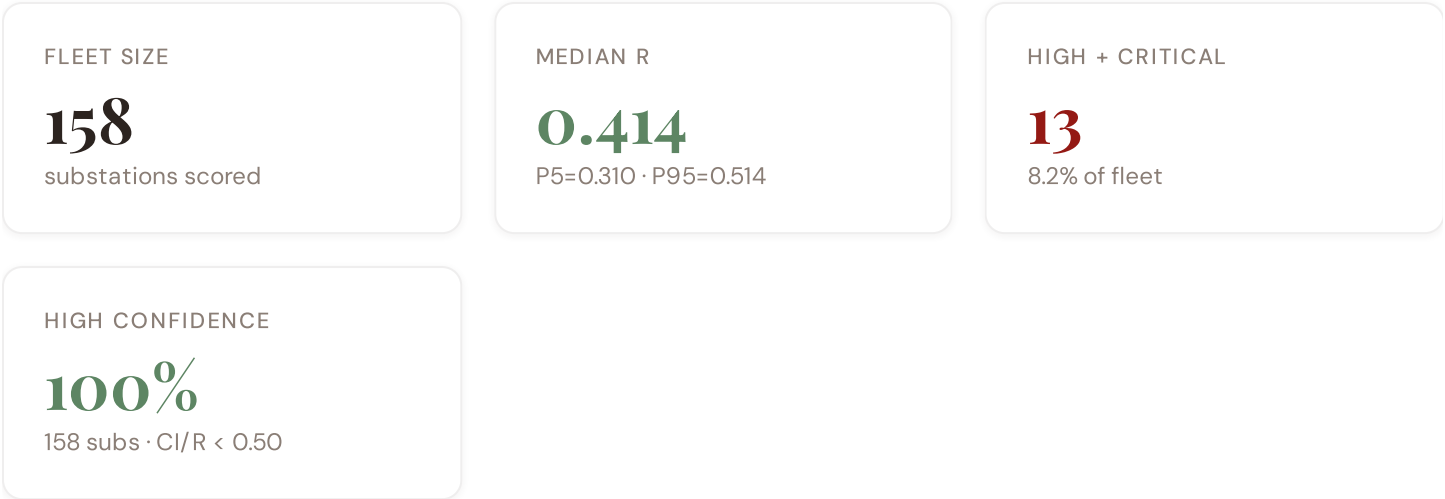
OSM Overpass · DRSV (floods) · Copernicus ERA5/CMIP6

Quarterly (5 sources)

DER registry + business + macro

DSO filings (5 Elektros) · SURS · Banka Slovenije · URSJV (Krško) · NEK Krško

C.2 Fleet Score Snapshot



No primary data sources have been updated since the v4.0.2 launch baseline. All **158 substation scores remain unchanged** this edition. The fleet median R of 0.414 (mean 0.413) reflects a distribution skewed toward the lower bands, with 0.0% in Low and 91.8% in Medium. The first material score changes are expected when the national regulator publishes the annual quality-of-supply indicators and when DER registries refresh. High-frequency sources (OSM, weather) update weekly but feed only infrastructure topology (R4) and thermal proxy (I5), which have minimal impact on fleet-level score distribution.

C.3 Changelog — v3.4 → v4.0.2

9 changes tracked across PO, PI, and P2 releases

SI-S25-9	ENHANCED	France-parity rebuild — 8 pages rewritten to match FR structure verbatim	KB §64
SI-S25-8	NEW	Section G deepDives — 12 NUTS-3 Slovenian regions (Savinjska + 11)	intelligence G
SI-S25-7	ENHANCED	Edition patcher anchored at FIRST_REFRESH 2026-07-09 (KB §58.6 padStart(2) + d>=21)	intelligence
SI-S25-6	NEW	ESG report — slovenia entry in COUNTRY_SOURCES (14 references)	esg-report
SI-S25-5	NEW	Methodology metadata extension — COMPONENTS/NORM/VALIDATION/CHANGELOG (FR parity)	metadata
SI-S25-4	NEW	R6 modifier set — seismic (Posavska/Goriška/Obalno-kraška) + flood (2023 Savinja anchor)	methodology
SI-S25-3	DATA	d05_osm LIVE — 158 substations via ISO3166-1=SI area filter	methodology
SI-S25-2	DATA	5 Elektro DSOs + ELES TSO + AGEN-RS regulator + URSJV nuclear safety wired	methodology

SECTION D — MONTHLY DEEP-DIVE

The Savinja Valley Corridor: Where Nuclear Reliance Meets Energy Transition Pressure

Deep-dive rotation: Each edition features a substantive thematic analysis. The 12-month rotation covers all 12 Slovenian NUTS-3 statistical regions: **Ed. 01** Savinja Valley & 2023 floods · **02** Ljubljana metropolitan · **03** Krško NPP zone (Posavska) · **04** Drava industrial (Podravska) · **05** Alpine valleys (Gorenjska) · **06** Adriatic coast (Obalno-kraška) · **07** Soča valley + Friuli seismic (Goriška) · **08** Krka industrial cluster (JV Slovenija) · **09** Pomurje energy poverty · **10** Koroška mountains · **11** Zasavje coal transition · **12** Karst plateau (Primorsko-notranjska).

D.1 Why the Savinja Valley, Why Now



The Savinja Valley hosts **18 substations** across 1 nuts-3 regions, making it the locus of this edition's deep-dive analysis. Its risk profile is concentrated in the upper bands: **5 substations (27.8%)** are classified High, with only **0** in the Low band. 78% of corridor substations score above the national fleet median of 0.414. The corridor median R of 0.483 reflects the local risk profile. The corridor median R reflects the tension between Krško NPP baseload inflexibility, rapidly growing renewable variability, and aging grid infrastructure built for 1970s load profiles. The SSI flags continuity-of-supply deficits, infrastructure age mismatches with modern demand patterns, and grid modernization lag under Slovenia Digital 2030 timelines.

D.2 Corridor Map and Scoring



SUBSTATION	PROVINCE	R_FINAL	BAND	C	V	I	E	S	T	ALERT
TP	SIO43	0.634	High	0.665	0.675	0.633	0.394	0.127	0.621	C, V
RP Dobravlje	SIO43	0.547	High	0.622	0.379	0.474	0.486	0.156	0.347	
RTP Kobarid	SIO43	0.532	High	0.424	0.650	0.470	0.475	0.168	0.595	V
RP Kanal	SIO43	0.522	High	0.579	0.298	0.595	0.521	0.169	0.451	
Stikališče Doblar	SIO43	0.517	High	0.586	0.576	0.562	0.563	0.169	0.612	
RTP Vrtojba	SIO43	0.499	Medium	0.610	0.383	0.573	0.591	0.165	0.563	
RTP Nova Gorica	SIO43	0.495	Medium	0.575	0.364	0.448	0.584	0.165	0.437	
RTP Ajdovščina	SIO43	0.485	Medium	0.547	0.632	0.552	0.476	0.121	0.356	
RTP Plave	SIO43	0.483	Medium	0.594	0.454	0.549	0.505	0.165	0.412	
RP Komen	SIO43	0.481	Medium	0.435	0.633	0.365	0.559	0.148	0.429	

... 8 additional substations — [explore full corridor in Digital Twin →](#)

D.3 Component Analysis

COMPONENT AVERAGES — SAVINJA VALLEY VS FLEET

C — Continuity	0.548 vs 0.485 (+13%)
I — Infrastructure	0.518 vs 0.481 (+8%)

MODIFIER AVERAGES — SAVINJA VALLEY VS FLEET

R3 Consequence	1.012	fleet 1.004 ↑
R4 Graph Crit.	0.970	fleet 0.977 ↓

S — Saturation	0.153 vs 0.109 (+40%)	R6a Restoration	1.045	fleet 1.020 ↑
V — Voltage	0.470 vs 0.472 (−0%)	R6b Seismic	1.060	fleet 1.039 ↑
E — Economic	0.503 vs 0.466 (+8%)	R7 Digital Read.	0.985	fleet 0.980 ↓
T — Transition	0.467 vs 0.465 (0%)			

The dominant risk driver across the Savinja Valley corridor is **T (Transition)**, averaging 0.467 against a fleet average of 0.465 — occupying 933% of its weight ceiling (w = 0.05). The second contributor is **E (Economic)** at 0.503 vs fleet 0.466 (503% of ceiling). Among modifiers, the R3 consequence multiplier averages 1.012 (fleet: 1.004), reflecting the socio-economic exposure of the corridor. The R6b seismic modifier averages 1.060, capturing the local seismic exposure — a dimension invisible to every competing framework.

D.4 NUTS-3 Region Breakdown

NUTS-3 RISK PROFILE — SORTED BY MEAN R

Longer bar = lower R = better resilience.



The nuts-3 regions-level breakdown reveals significant intra-regional variation. **SIO43** leads with a mean R of 0.478 across 18 substations, while **SIO43** scores 0.478 — a spread of 0.000 within a single corridor. The SSI's substation-level granularity reveals that the Savinja Valley is not uniformly stressed but contains distinct sub-corridors of concentrated risk — precisely the kind of spatial pattern that investment planning requires.

D.5 Implications

O Savinja Valley substations score R ≥ 0.650, placing them in the upper quartile of national risk. For NEPN / national digital plan / RRP investment prioritisation, this analysis identifies the SIO43 sub-corridor as the highest-priority target — where DER deployment is accelerating against ageing infrastructure with above-average continuity deficits and significant socio-economic exposure. The SSI's silo-breaking approach reveals what no single-discipline framework can: that these substations matter not only because of their engineering condition, but because the communities they serve are disproportionately vulnerable to disruption. This is precisely the anticipatory grid planning capability that the SSI was built to provide.

SECTION E — RECURRING MODULE

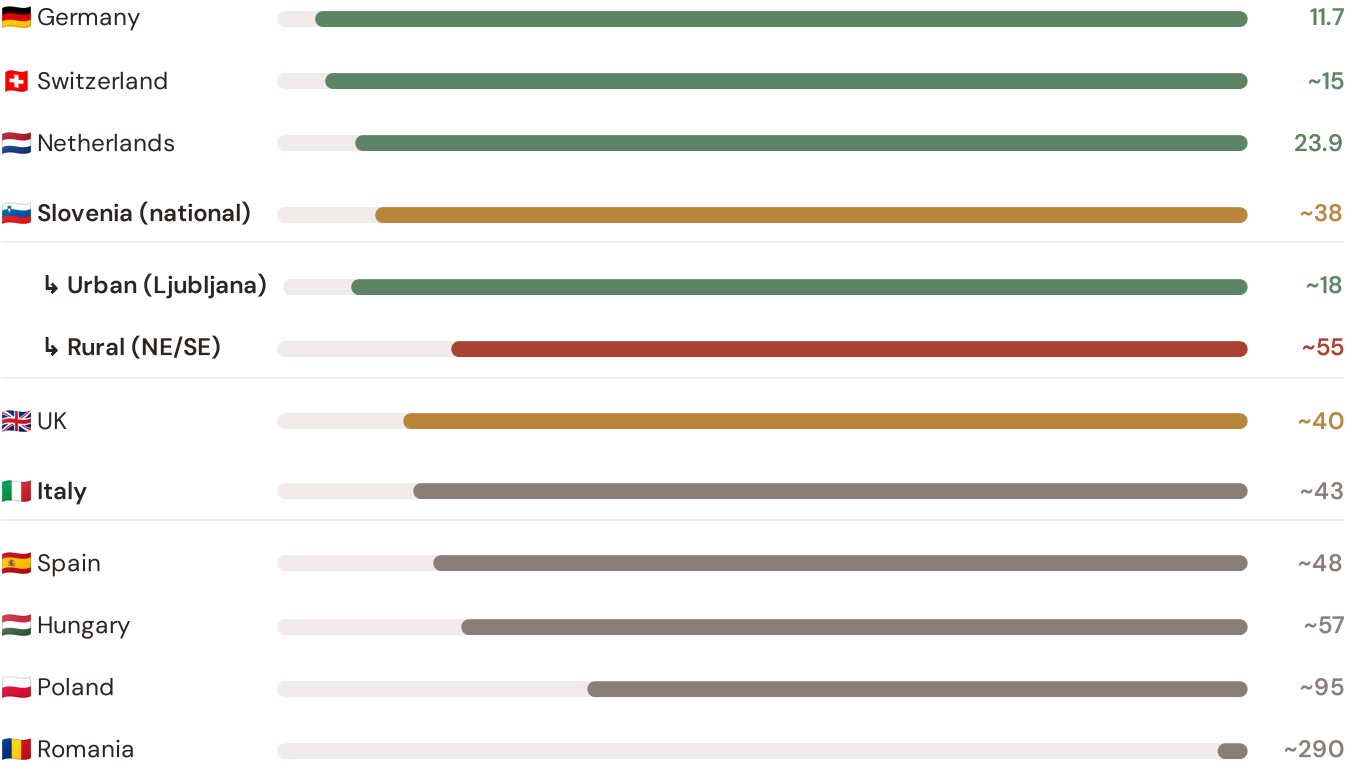
European Context Card

How this works:

Each edition includes one European Context Card drawn from the §16 Peer Benchmarking framework. The card is selected to complement the month's deep-dive theme. This month: **SAIDI comparison** — because the Savinja Valley corridor analysis hinges on Slovenia's continuity performance relative to European peers. Future editions rotate through: VoLL comparison (Ed. 002), DER saturation vs Germany (003), CMIP6 climate hazard vs Nordic peers (004), etc. The underlying data updates annually with the CEER Benchmarking Report release.

Unplanned SAIDI (min/customer/year) — Selected Countries

Longer bar = lower SAIDI = better reliability. Values in min/customer/year.



Source: CEER 7th Benchmarking Report (2022); AGEN-RS quality reports (2023); AGEN-RS (2024). Slovenia sub-categories from AGEN-RS territory classification (urban/rural). · See §16 Peer Benchmarking for full methodology and caveats. · Updated annually with CEER BR release.

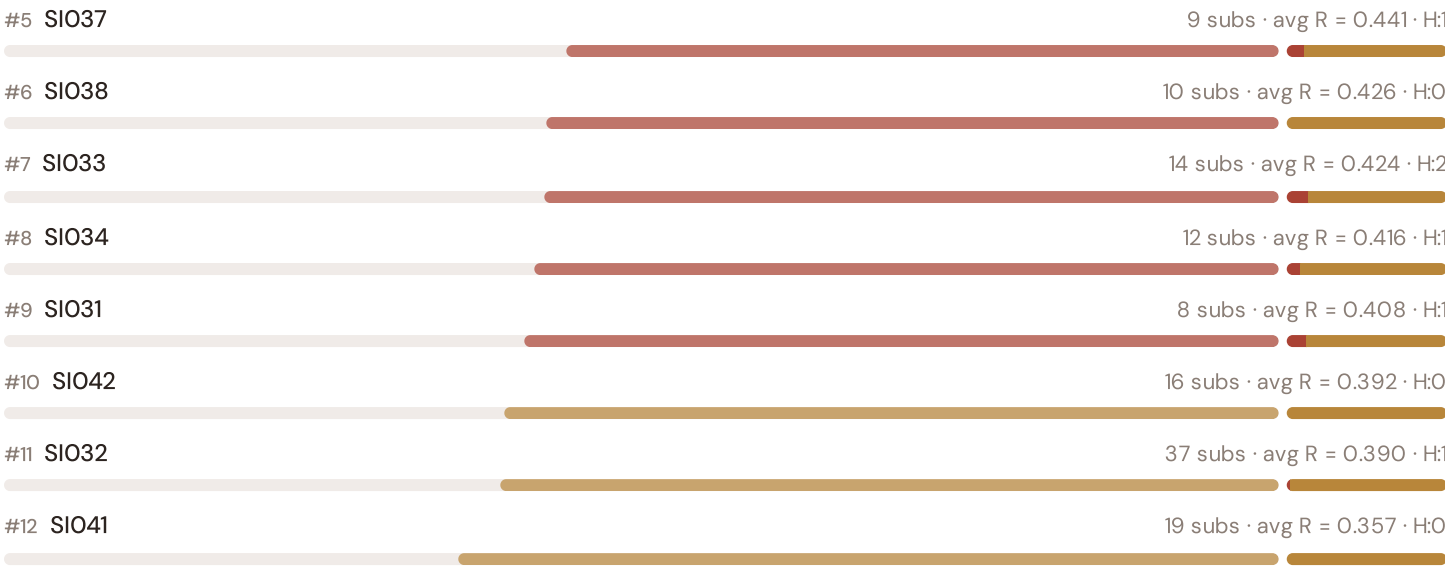
This month's key insight: The Savinja Valley corridor deep-dive shows **0 substations** (of 0) with C-component above 70% of its weight ceiling — indicating continuity-of-supply performance consistent with rural territory classification. The corridor's average C of 0.000 is **0.0× the fleet average** of 0.485. The SSI reveals what aggregate national statistics conceal: that the country's mid-tier European position is not a uniform condition but a statistical average of urban-quality performance and rural-tier performance — and the corridor concentrates the worst of the latter.

Regional Risk Ranking — All 13 Slovenian NUTS-3 Regions

Where does the Savinja Valley sit within the national landscape? Regions sorted by mean R_median, with band distribution and fleet comparison.

Longer bar = lower R = better resilience. Sorted worst → best.



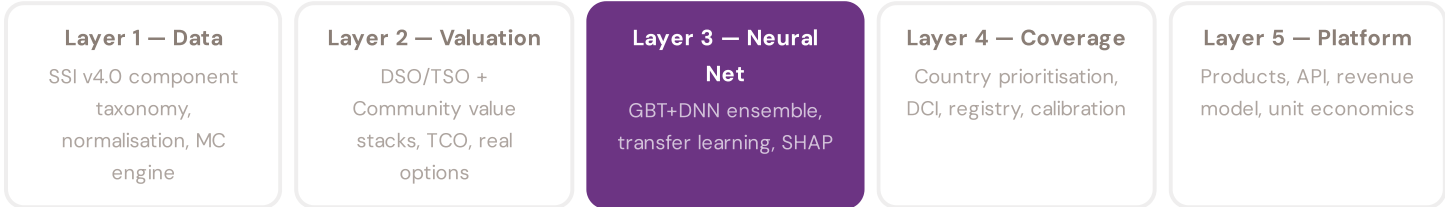


The Savinja Valley ranks **#0 of 12 nuts-3 regions** by mean R_median, placing it among the upper tier of national risk. The three most stressed are SIO43, SIO44, SIO36, while the three most resilient are SIO41, SIO32, SIO42. 8 of 12 score above the fleet average of 0.413. This ranking — possible only because the SSI scores every substation individually rather than relying on aggregate DSO or national statistics — reveals the true spatial distribution of grid stress. It is precisely this substation-level granularity that positions the SSI as a tool for investment prioritisation: not treating entire regions as monoliths, but identifying the specific corridors within each that demand attention.

SECTION F — RECURRING MODULE

SSI Enhanced Neural Network — Monthly Topic

From grid intelligence to BESS valuation platform. The SSI Index answers *"How healthy is this substation?"* The SSI Enhanced Neural Network (SSI-ENN) extends this into: *"What is a BESS worth at this substation — to the Elektro DSOs/ELES and to the community?"* Each edition of this report explores one building block of the SSI-ENN's five-layer architecture, showing how the SSI Index data feeds directly into investment-grade grid analytics.



LAYER 3 §0.5.6 · June 2026

Explainability & SHAP Values

Clients need to understand not just the prediction but WHY a substation ranks highly

Investment-grade models require interpretability. SHAP (SHapley Additive exPlanations) provides a rigorous, game-theoretic attribution of each feature's contribution to each prediction. For every substation, the SSI-ENN produces a SHAP waterfall: starting from the fleet-average valuation and showing how each feature pushes the estimate up or down. This directly connects to the SSI Index: if SHAP identifies C (Continuity) as the dominant positive driver, the client knows the

BESS value depends primarily on the substation's poor reliability history. Transparency converts model outputs into investment decisions.

KEY CONCEPTS

→ SHAP: game-theoretic feature attribution per prediction

→ Waterfall: fleet average → substation prediction

→ Direct traceability to SSI components and data sources

→ Feature importance validated against domain expertise

LIVE SSI DATA CONNECTION

The training set comprises 158 substations with complete feature vectors. Average CI_width of 0.074 across the fleet provides the uncertainty ground truth that the neural network's quantile regression heads must calibrate against. The fleet's risk distribution — 0 Low, 145 Medium, 13 High, 0 Critical — ensures balanced training across all risk bands.

Coming next month:

LAYER 3

 Uncertainty Quantification — *Investment-grade predictions require calibrated confidence intervals, not just point estimates*

SECTION G

Looking Ahead

NEXT MONTH — EDITION 02

The Koroška Mountains

Deep-dive into Koroška's grid risk profile — substation map, component analysis, nuts-3 regions breakdown. European Context Card: Light-rural tier — limited substation density, AT cross-border dependencies.

NEXT DATA REFRESH

Q3 2026 — 5 Quarterly Sources

DSO filings (5 Elektros), SURS, Banka Slovenije, URSJV (Krško), NEK Krško are due for refresh in Q3. Impact: DER registry (T1), business fabric indicators (E2), macro-economic data. Highest-impact annual source pending: **AGEN-RS — Agencija za energijo** (12 variables → C1,C2,V1,V2,T1).

FLEET SPOTLIGHT

SIO43: Highest Risk Concentration

SIO43 has the highest concentration of High/Critical substations: **5 of 18 (28%)** are in the upper risk bands — avg R = 0.477. This makes it the most acute risk corridor in the fleet.

Edition 02 will be published on the second Thursday of July 2026. Deep-dive region: **Koroška**. To ensure you receive it, confirm your subscription segment (General / Institutional / Academic) by email to ssi_index@ikenga.eu.

Feedback welcome. The SSI Index is designed to evolve through stakeholder dialogue. If you represent a Slovenian utility, AGEN-RS, ELES, Elektro DSOs, municipal energy company, or academic institution and would like to contribute data or methodology feedback, please contact us at ssi_index@ikenga.eu. The SSI's credibility depends on open scrutiny.

SSI Monthly Intelligence Report — Slovenia Edition 01 (Inaugural)

Published 9 July 2026 · SSI Index v4.0.2 · 158 substations · ~860 grid lines · 6 components · 20 metrics · 6 modifiers
10,000-iteration Gaussian Copula Monte Carlo · Sobol global sensitivity analysis (196,608 evals/substation) · CMIP6 SSP2-4.5